

Full-stack software engineer working end-to-end across web, backend, and ML; experience includes startups, Google, Amazon, and Meta. A particular passion of mine is using machine learning to answer questions and surface information that would otherwise be too difficult to communicate. I believe that we are quickly moving towards a time where the boundary between user and computer will fade and I want to help us get there.

Meta — Software Engineer, Machine Learning Generalist

2024 — now · python · Hack · pytorch

- ML engineer on Instagram's Relevance Labs team, building and shipping new recommendation features.
- Worked on the initial launch of Friendly Viewer (shows which of your friends liked a post with little profile bubbles).
- Brought Blend through global launch (combine your algorithm with a friend's in a DM thread).
- Initial engineer on Your Algorithm; built the prototype with two other coworkers and currently own "see less" (users tell the algorithm which topics they want to see more or less of).

Amazon — Software Engineer, Robotics AI

2022 — 2023 · python · pytorch · AWS

- Initially worked on computer vision for warehouse robotics.
- Completed a camera calibration project before transitioning to a damage detection team due to team merger.
- Delivered detailed analyses and trained models for integration into the final Mixture of Experts (MoE) model.
- Contributed a model instrumental in achieving the team's year-end accuracy goal for the MoE system.

Google — Dialer Team Software Engineer

2019 — 2022 · java · kotlin · android

- Co-inventor of the patent for automatic call screening, improving Android's default Phone app.
- Collaborated with a large team to build the automatic call screening feature.
- Worked with one other coworker to prototype a new test framework that is now in use to run the same test suite as unit/emulator/device tests.
- Handled client side implementation of the initial i18n pilot for call screen (Japanese support).
- Implemented and launched caller tags, a less specific type of caller ID that tells users what type of business is calling.
- Set up crowdsourcing flows to continually improve our caller ID/tag coverage.
- Built server side pipeline for internationalizing and hosting the set of available caller tags. Continuing work on server side heuristics/anti abuse policy to choose caller tags based on past feedback.

Tophatter — Full Stack Software Engineer

2017 — 2018 · rails · coffeescript · python · spark · tensorflow

- Spent a hack week on web performance; implemented viewport rendering and improved mobile web performance, producing a stat-sig +6% new-user conversion and +4% total web revenue.
- Learned Rails and transitioned to full stack.
- Pitched and worked with a team of 3 to build a collaborative filtering model for product recommendation.
- Worked with team of 3 to rebuild product deduplication pipeline from the ground up. Used neural networks to process item features into a few feature vectors, generated candidates using faiss vector similarity search, judged if candidates were actually duplicates using a random forest.
- Pitched, designed, and built a human review queue for our model to get ground truth data for product deduplication. This improved the performance of a NN I had set aside before to the point where it replaced the random forest model. Closed the training loop so the system will continue to improve with time.
- Infrastructure work on data pipelines. Built service that efficiently tracks recent user-product interactions using ring buffers in redis.

Pandora — Web Team Engineer

2015 — 2017 · JS (react) · HTML · CSS

- Worked on angular UI for a tool for artists to send messages to their listeners.
- Spent last year and a half working on complete react rewrite of web client.
- Implemented Pandora Premium subscription tier on web.

Rdio — Web Team Engineer

2014 — 2015 · JS (backbone) (react) · HTML · CSS · SQL · python

- Worked on "music feed", the final rdio.com homepage page.
- Pitched, created, and deployed a viewport tracker for the content on our homepage.
- Implemented Rdio Select subscription tier on web.

UIUC

2011 — 2014

- Studied Materials Science and Computer Science.
- Eclectic class list.
- 2nd place in Microsoft Windows Phone hackathon. I won a copy of viva pinata.
- Participated in Engineering Open House the last 2 years I was there.

St. Louis Priory School

2006 — 2011

- Tutored Chemistry and Physics for school program. Then proceeded to start much more lucrative independent classroom-style prep courses for Chemistry that I ran for the summers of 2011-12.
- National Merit Semi-Finalist
- FIRST robotics team 1329 leader